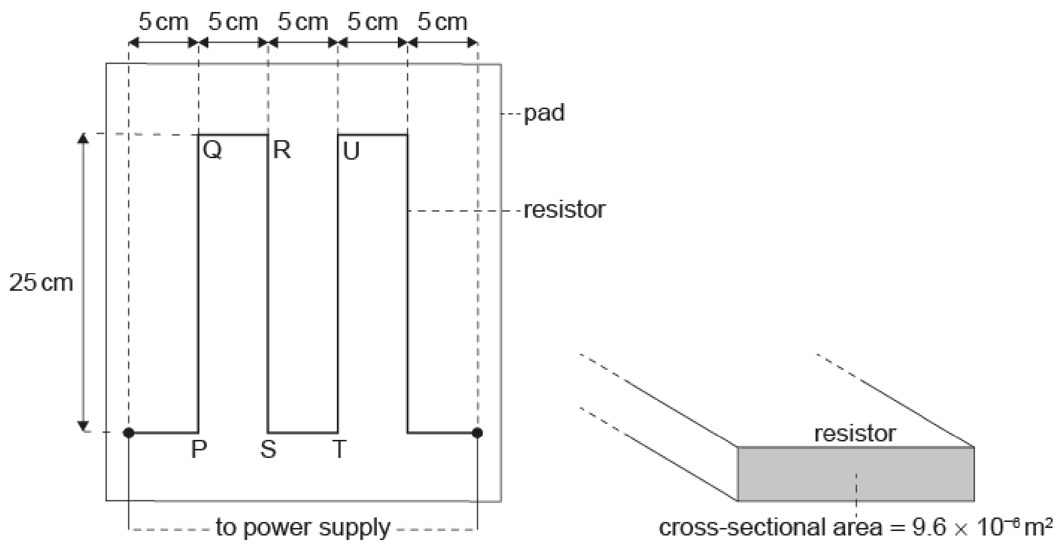
Study Guide - SL - Topic B5 D2 D3 D4 - Paper 2 [125 marks]

1. [Maximum mark: 12] 23M.2.SL.TZ2.4

An electrically heated pad is designed to keep a pet warm.

The pad is heated using a resistor that is placed inside the pad. The dimensions of the resistor are shown on the diagram. The resistor has a resistance of 4.2 Ω and a total length of 1.25 m.

diagram not to scale



When there is a current in the resistor, the temperature in the pad changes from a room temperature of 20 °C to its operating temperature at 35 °C.

(a) The designers state that the energy transferred by the resistor every second is 15 J.

Calculate the current in the resistor. [1]

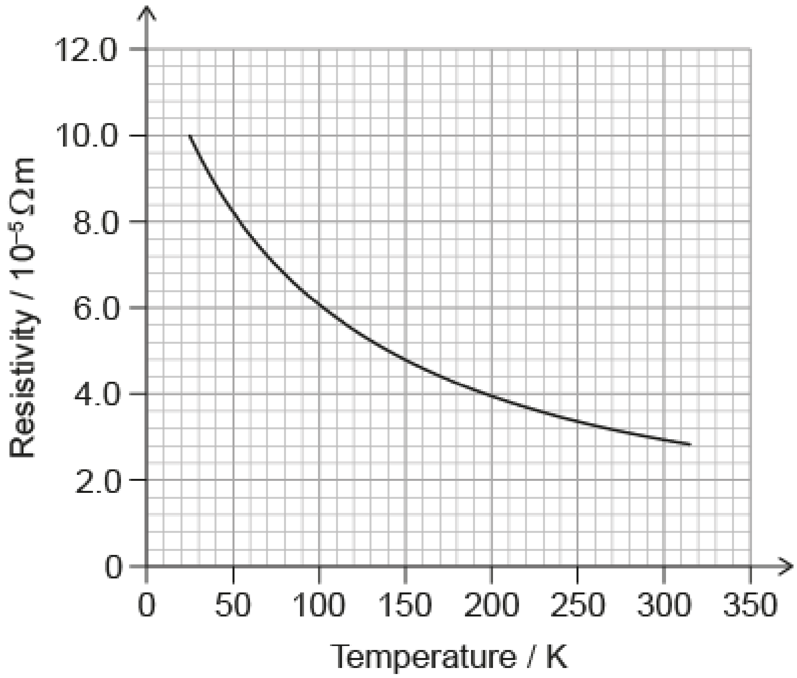
. . . .

. . . .

. . . .

(b) The designers wish to make the resistor from carbon fibre.

The graph shows the variation with temperature, in Kelvin, of the resistivity of carbon fibre.



(b.i) The resistor has a cross-sectional area of 9.6×10^{-6} m^2.

Show that a resistor made from carbon fibre will be suitable for the pad. [3]

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

(b.ii) The power supply to the pad has a negligible internal resistance.

State and explain the variation in current in the resistor as the temperature of the pad increases. [2]

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

(c) When there is a current in the resistor, magnetic forces act between the resistor strips.

For the part of the resistor labelled RS,

(c.i) outline the magnetic force acting on it due to the current in PQ. [1]

. . . .

. . . .

. . . .

(c.ii) state and explain the net magnetic force acting on it due to the currents in PQ and TU. [2]

. . . .

. . . .

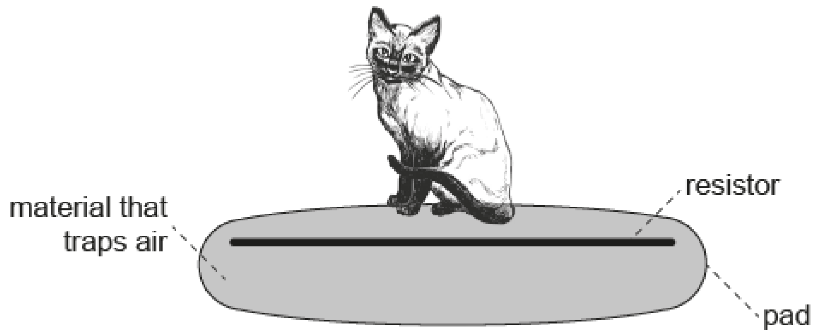
. . . .

. . . .

. . . .

. . . .

(d) The design of the pad encloses the resistor in a material that traps air. The design also places the resistor close to the top surface of the pad.



[3]

Explain, with reference to thermal energy transfer, why the pad is designed in this way.

....

....

....

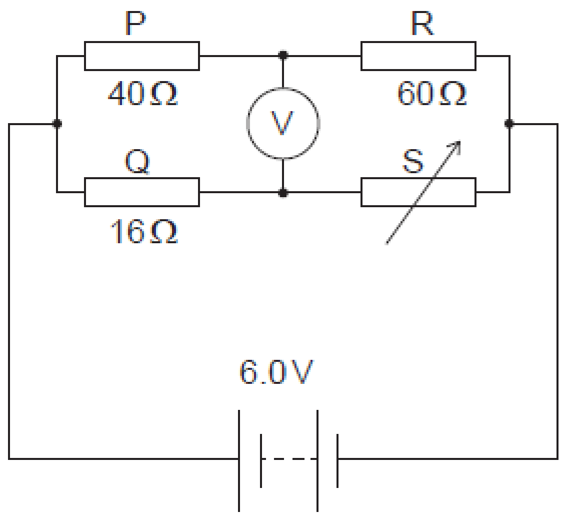
....

....

....

2. [Maximum mark: 10] 22M.2.SL.TZ1.4

A power supply is connected to three resistors P, Q and R of fixed value and to an ideal voltmeter. A variable resistor S, formed from a solid cylinder of conducting putty, is also connected in the circuit. Conducting putty is a material that can be moulded so that the resistance of S can be changed by re-shaping it.



The resistance values of P, Q and R are 40 Ω, 16 Ω and 60 Ω respectively. The emf of the power supply is 6.0 V and its internal resistance is negligible.

(a) Calculate the potential difference across P. [2]

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

(b) The voltmeter reads zero. Determine the resistance of S. [3]

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

All the putty is reshaped into a solid cylinder that is four times longer than the original length.

(c.i) Deduce the resistance of this new cylinder when it has been reshaped. [3]

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

(c.ii) Outline, without calculation, the change in the total power dissipated in Q and the new cylinder after it has been reshaped. [2]

3. [Maximum mark: 14] 22M.2.SL.TZ2.4

(a) Identify the laws of conservation that are represented by Kirchhoff's circuit laws. [2]

. . . .

. . . .

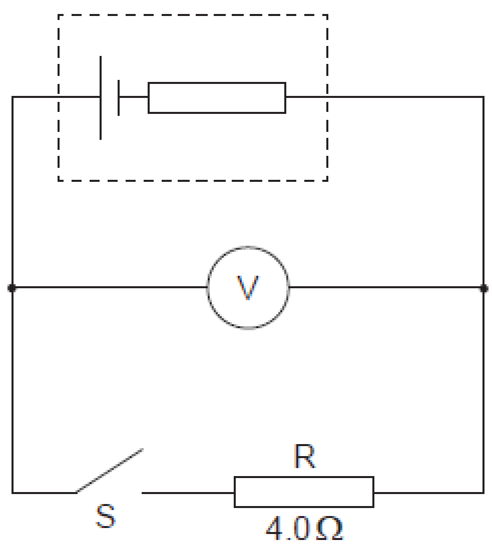
. . . .

. . . .

. . . .

. . . .

A cell is connected to an ideal voltmeter, a switch S and a resistor R. The resistance of R is 4.0 Ω.



When S is open the reading on the voltmeter is 12 V. When S is closed the voltmeter reads 8.0 V.

(b.i) State the emf of the cell. [1]

. . . .

. . . .

. . . .

(b.ii) Deduce the internal resistance of the cell. [2]

. . . .

. . . .

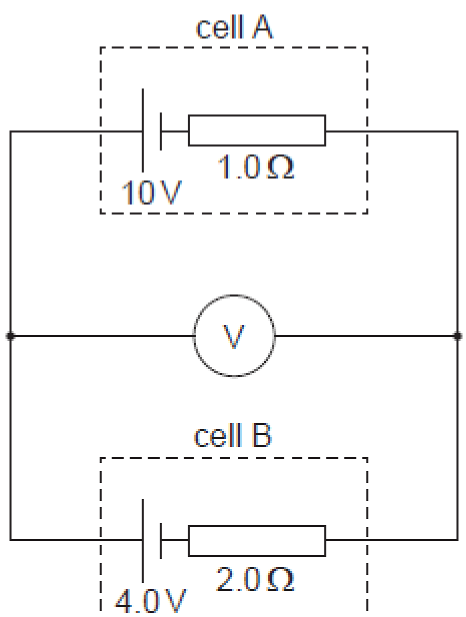
. . . .

. . . .

. . . .

. . . .

(c) The voltmeter is used in another circuit that contains two secondary cells.



Cell A has an emf of 10 V and an internal resistance of 1.0 Ω. Cell B has an emf of 4.0 V and an internal resistance of 2.0 Ω.

Calculate the reading on the voltmeter. [3]

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

Electricity can be generated using renewable resources.

(d.i) Outline why electricity is a secondary energy source. [1]

. . . .

. . . .

. . . .

(d.ii) Some fuel sources are renewable. Outline what is meant by renewable. [1]

. . . .

. . . .

. . . .

(e.i) A fully charged cell of emf 6.0 V delivers a constant current of 5.0 A for a time of 0.25 hour until it is completely discharged.

The cell is then re-charged by a rectangular solar panel of dimensions 0.40 m $\times$ 0.15 m at a place where the maximum intensity of sunlight is 380 W m^{-2}.

The overall efficiency of the re-charging process is 18 %.

Calculate the minimum time required to re-charge the cell fully. [3]

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

(e.ii) Outline why research into solar cell technology is important to society. [1]

. . . .

. . . .

. . . .

4. [Maximum mark: 5] 21M.2.SL.TZ1.2

A planet is in a circular orbit around a star. The speed of the planet is constant.

(a.i) Explain why a centripetal force is needed for the planet to be in a circular orbit. [2]

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

(a.ii) State the nature of this centripetal force. [1]

. . . .

. . . .

. . . .

(b) Determine the gravitational field of the planet.

The following data are given:

Mass of planet $= 8.0 \times 10^{24}$ kg
Radius of the planet $= 9.1 \times 10^{6}$ m. [2]

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

5. [Maximum mark: 12] 21M.2.SL.TZ1.3

A mass of 1.0 kg of water is brought to its boiling point of 100 °C using an electric heater of power 1.6 kW.

(a.i) The molar mass of water is 18 g mol^{-1}. Estimate the average speed of the water molecules in the vapor produced. Assume the vapor behaves as an ideal gas. [2]

....

....

....

....

....

....

(a.ii) State **one** assumption of the kinetic model of an ideal gas. [1]

....

....

....

A mass of 0.86 kg of water remains after it has boiled for 200 s.

(b.i) Estimate the specific latent heat of vaporization of water. State an appropriate unit for your answer. [2]

....

....

....

....

....

....

(b.ii) Explain why the temperature of water remains at 100 °C during this time. [1]

....

....

....

(c) The heater is removed and a mass of 0.30 kg of pasta at −10 °C is added to the boiling water.

Determine the equilibrium temperature of the pasta and water after the pasta is added. Other heat transfers are negligible.

Specific heat capacity of pasta = 1.8 kJ kg^{-1} K^{-1}
Specific heat capacity of water = 4.2 kJ kg^{-1} K^{-1} [3]

....

....

....

....

....

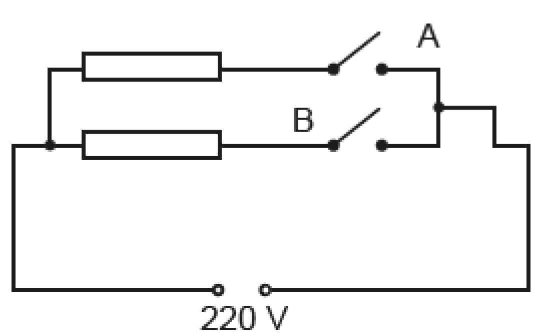
....

....

....

....

The electric heater has two identical resistors connected in parallel.



The circuit transfers 1.6 kW when switch A only is closed. The external voltage is 220 V.

(d.i) Show that each resistor has a resistance of about 30 Ω. [1]

....

....

....

(d.ii) Calculate the power transferred by the heater when both switches are closed. [2]

....

....

....

....

....

....

6. [Maximum mark: 10] 21M.2.SL.TZ2.6

A photovoltaic cell is supplying energy to an external circuit. The photovoltaic cell can be modelled as a practical electrical cell with internal resistance.

The intensity of solar radiation incident on the photovoltaic cell at a particular time is at a maximum for the place where the cell is positioned.

The following data are available for this particular time:

Operating current = 0.90 A
Output potential difference to external circuit = 14.5 V
Output emf of photovoltaic cell = 21.0 V
Area of panel = 350 mm × 450 mm

(a) Explain why the output potential difference to the external circuit and the output emf of the photovoltaic cell are different.

[2]

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

(b) Calculate the internal resistance of the photovoltaic cell for the maximum intensity condition using the model for the cell.

[3]

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

(c) The maximum intensity of sunlight incident on the photovoltaic cell at the place on the Earth's surface is 680 W m^{-2}.

A measure of the efficiency of a photovoltaic cell is the ratio

$$\frac{\text{energy available every second to the external circuit}}{\text{energy arriving every second at the photovoltaic cell surface}}.$$

Determine the efficiency of this photovoltaic cell when the intensity incident upon it is at a maximum. [3]

. . . .

. . . .

. . . .

(d) State **two** reasons why future energy demands will be increasingly reliant on sources such as photovoltaic cells.

Reason 1: . . . .

. . . .

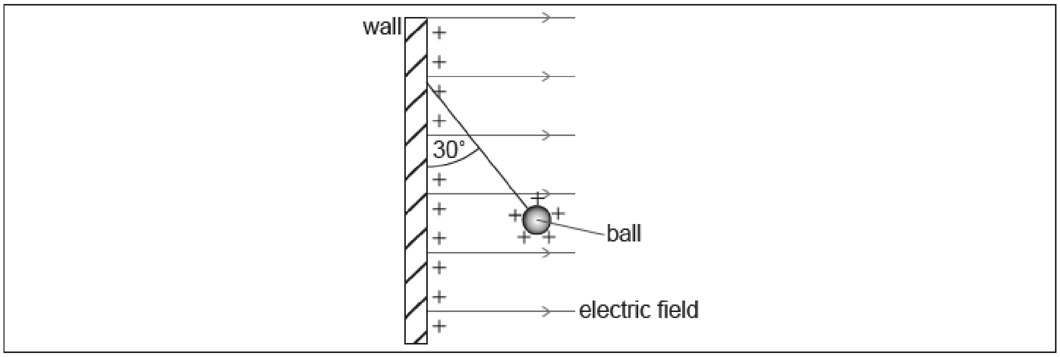
Reason 2: . . . .

. . . .

[2]

7. [Maximum mark: 9] 21M.2.SL.TZ2.3

A vertical wall carries a uniform positive charge on its surface. This produces a uniform horizontal electric field perpendicular to the wall. A small, positively-charged ball is suspended in equilibrium from the vertical wall by a thread of negligible mass.



(a) The charge per unit area on the surface of the wall is σ. It can be shown that the electric field strength E due to the charge on the wall is given by the equation

$$E = \frac{\sigma}{2\varepsilon_0}.$$

Demonstrate that the units of the quantities in this equation are consistent. [2]

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

(b.i) The thread makes an angle of 30° with the vertical wall. The ball has a mass of 0.025 kg.

Determine the horizontal force that acts on the ball. [3]

....

....

....

....

....

....

(b.ii) The charge on the ball is 1.2×10^{-6} C. Determine σ. [2]

....

....

....

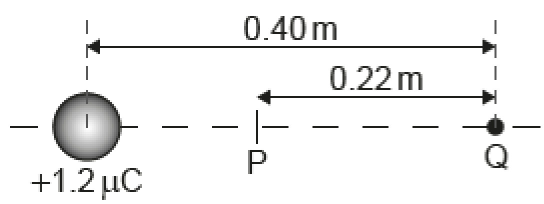
....

....

....

(c) The centre of the ball, still carrying a charge of $1.2 \times 10^{-6}\,\mathrm{C}$, is now placed $0.40\,\mathrm{m}$ from a point charge Q. The charge on the ball acts as a point charge at the centre of the ball.

P is the point on the line joining the charges where the electric field strength is zero.
The distance PQ is $0.22\,\mathrm{m}$.



Calculate the charge on Q. State your answer to an appropriate number of significant figures. [2]

8. [Maximum mark: 15] 19M.2.SL.TZ1.1

A girl rides a bicycle that is powered by an electric motor. A battery transfers energy to the electric motor. The emf of the battery is 16 V and it can deliver a charge of 43 kC when discharging completely from a full charge.

The maximum speed of the girl on a horizontal road is 7.0 m s^{-1} with energy from the battery alone. The maximum distance that the girl can travel under these conditions is 20 km.

(a.i) Show that the time taken for the battery to discharge is about 3 $\times 10^3$ s. [1]

. . . .

. . . .

. . . .

(a.ii) Deduce that the average power output of the battery is about 240 W. [2]

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

(a.iii) Friction and air resistance act on the bicycle and the girl when they move. Assume that all the energy is transferred from the battery to the electric motor. Determine the total average resistive force that acts on the bicycle and the girl. [2]

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

The bicycle and the girl have a total mass of 66 kg. The girl rides up a slope that is at an angle of 3.0° to the horizontal.

(b.i) Calculate the component of weight for the bicycle and girl acting down the slope. [1]

. . . .

. . . .

. . . .

(b.ii) The battery continues to give an output power of 240 W. Assume that the resistive forces are the same as in (a)(iii).

Calculate the maximum speed of the bicycle and the girl up the slope. [2]

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

(c) On another journey up the slope, the girl carries an additional mass. Explain whether carrying this mass will change the maximum distance that the bicycle can travel along the slope. [2]

. . . .

. . . .

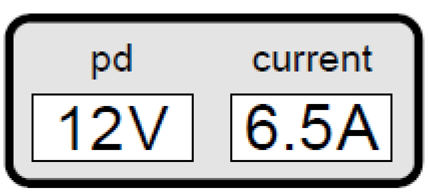
. . . .

. . . .

. . . .

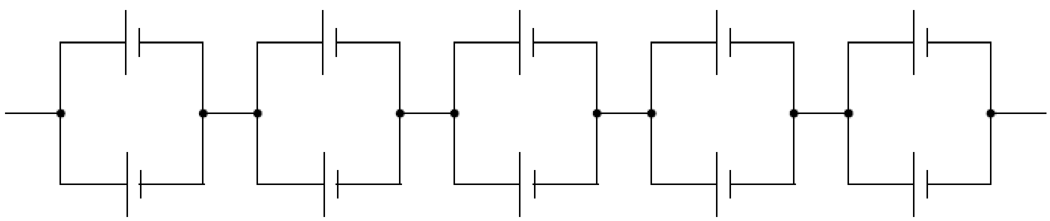
. . . .

The bicycle has a meter that displays the current and the terminal potential difference (pd) for the battery when the motor is running. The diagram shows the meter readings at one instant. The emf of the cell is 16 V.



(d) Determine the internal resistance of the battery. [2]

The battery is made from an arrangement of 10 identical cells as shown.

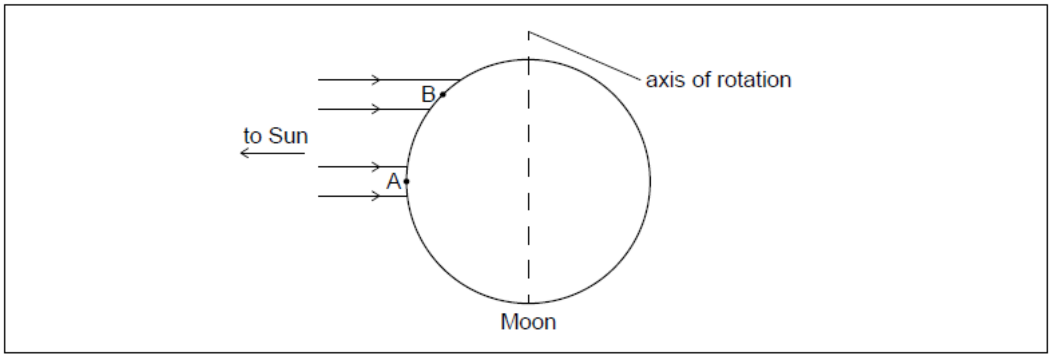


(e.i) Calculate the emf of **one** cell. [1]

(e.ii) Calculate the internal resistance of **one** cell. [2]

9. [Maximum mark: 7] 19M.2.SL.TZ1.6

The Moon has no atmosphere and orbits the Earth. The diagram shows the Moon with rays of light from the Sun that are incident at 90° to the axis of rotation of the Moon.



(a.i) A black body is on the Moon's surface at point A. Show that the maximum temperature that this body can reach is 400 K. Assume that the Earth and the Moon are the same distance from the Sun. [2]

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

(a.ii) Another black body is on the Moon's surface at point B.

Outline, without calculation, why the aximum temperature of the black body at point B is less than at point A. [2]

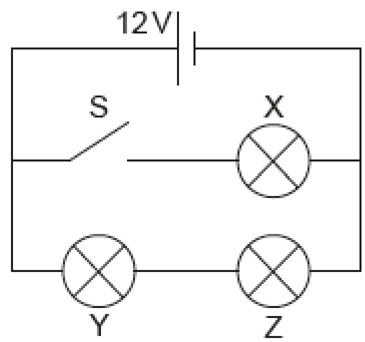
(b) The albedo of the Earth's atmosphere is 0.28. Outline why the maximum temperature of a black body on the Earth when the Sun is overhead is less than that at point A on the Moon. [1]

(c.i) Outline why a force acts on the Moon. [1]

(c.ii) Outline why this force does no work on the Moon. [1]

10. [Maximum mark: 5] 19M.2.SL.TZ2.4

Three identical light bulbs, X, Y and Z, each of resistance 4.0 Ω are connected to a cell of emf 12 V. The cell has negligible internal resistance.



(a) The switch S is initially open. Calculate the total power dissipated in the circuit. [2]

. . . .

. . . .

. . . .

(bi) The switch is now closed. State, without calculation, why the current in the cell will increase. [1]

. . . .

. . . .

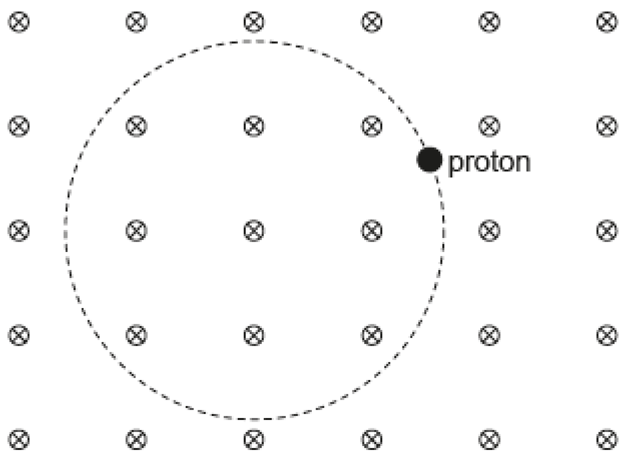
. . . .

(bii) The switch is now closed. Deduce the ratio $\frac{\text{power dissipated in Y with S open}}{\text{power dissipated in Y with S closed}}$.

[2]

11. [Maximum mark: 5] 19M.2.SL.TZ2.5

A proton moves along a circular path in a region of a uniform magnetic field. The magnetic field is directed into the plane of the page.



(ai) Label with arrows on the diagram the magnetic force *F* on the proton. [1]

. . . .

. . . .

. . . .

(aii) Label with arrows on the velocity vector *v* of the proton. [1]

. . . .

. . . .

. . . .

(b) The speed of the proton is 2.16×10^6 m s^{-1} and the magnetic field strength is 0.042 T. For this proton, determine, in m, the radius of the circular path. Give your answer to an appropriate number of significant figures. [3]

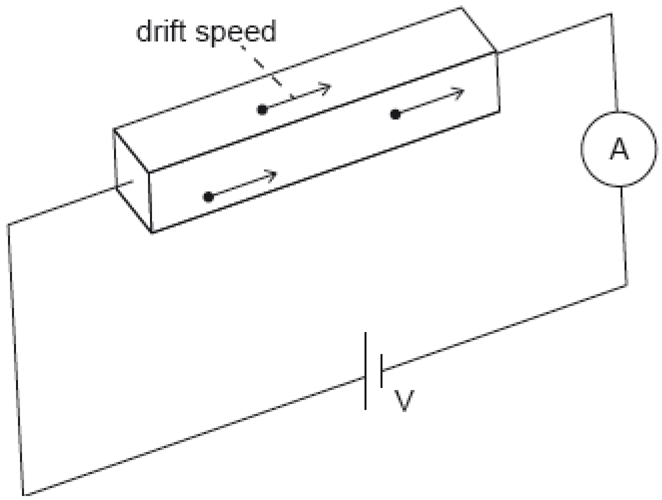
. . . .

. . . .

. . . .

12. [Maximum mark: 5] 18M.2.SL.TZ1.4

An ohmic conductor is connected to an ideal ammeter and to a power supply of output voltage V.



The following data are available for the conductor:

density of free electrons $= 8.5 \times 10^{22}\ \text{cm}^{-3}$

resistivity $\rho = 1.7 \times 10^{-8}\ \Omega\text{m}$

dimensions $w \times h \times l = 0.020\ \text{cm} \times 0.020\ \text{cm} \times 10\ \text{cm}$.

The ammeter reading is 2.0 A.

(a) Calculate the resistance of the conductor. [2]

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

(b) Calculate the drift speed v of the electrons in the conductor in cm s^{-1}. State your answer to an appropriate number of significant figures. [3]

....

....

....

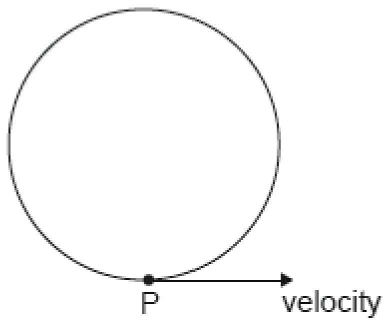
....

....

....

13. [Maximum mark: 5] 18M.2.SL.TZ1.5

An electron moves in circular motion in a uniform magnetic field.



The velocity of the electron at point P is 6.8×10^{5} m s^{-1} in the direction shown.

The magnitude of the magnetic field is 8.5 T.

(a) State the direction of the magnetic field. [1]

. . . .

. . . .

. . . .

(b) Calculate, in N, the magnitude of the magnetic force acting on the electron. [1]

. . . .

. . . .

. . . .

(c.i) Explain why the electron moves at constant speed. [1]

. . . .

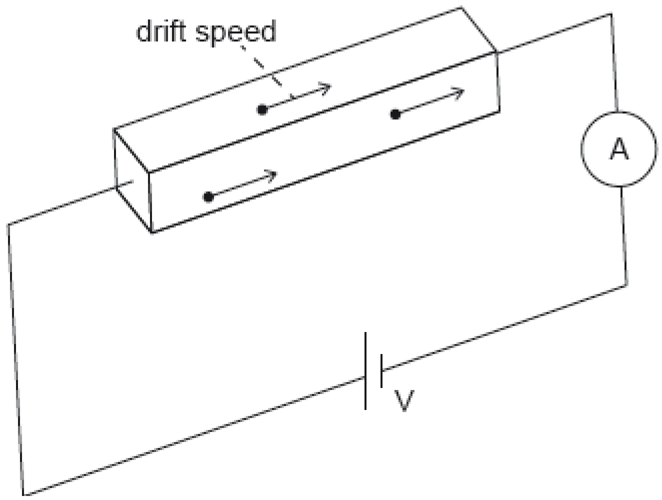
. . . .

. . . .

(c.ii) Explain why the electron moves on a circular path. [2]

14. [Maximum mark: 5] 18M.2.SL.TZ1.4

An ohmic conductor is connected to an ideal ammeter and to a power supply of output voltage V.



The following data are available for the conductor:

density of free electrons $= 8.5 \times 10^{22}\ \text{cm}^{-3}$

resistivity $\rho = 1.7 \times 10^{-8}\ \Omega\text{m}$

dimensions $w \times h \times l = 0.020\ \text{cm} \times 0.020\ \text{cm} \times 10\ \text{cm}$.

The ammeter reading is 2.0 A.

(a) Calculate the resistance of the conductor. [2]

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

(b) Calculate the drift speed v of the electrons in the conductor in $cm\ s^{-1}$. State your answer to an appropriate number of significant figures. [3]

....

....

....

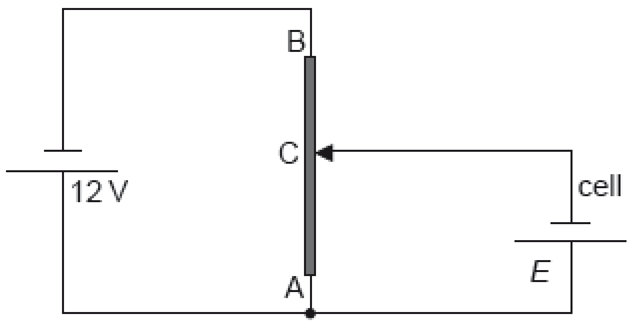
....

....

....

15. [Maximum mark: 6] 18M.2.SL.TZ2.4

The diagram shows a potential divider circuit used to measure the emf *E* of a cell X. Both cells have negligible internal resistance.



(a) State what is meant by the emf of a cell. [2]

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

AB is a wire of uniform cross-section and length 1.0 m. The resistance of wire AB is 80 Ω. When the length of AC is 0.35 m the current in cell X is zero.

(b.i) Show that the resistance of the wire AC is 28 Ω. [2]

. . . .

. . . .

. . . .

. . . .

. . . .

. . . .

(b.ii) Determine *E*. [2]